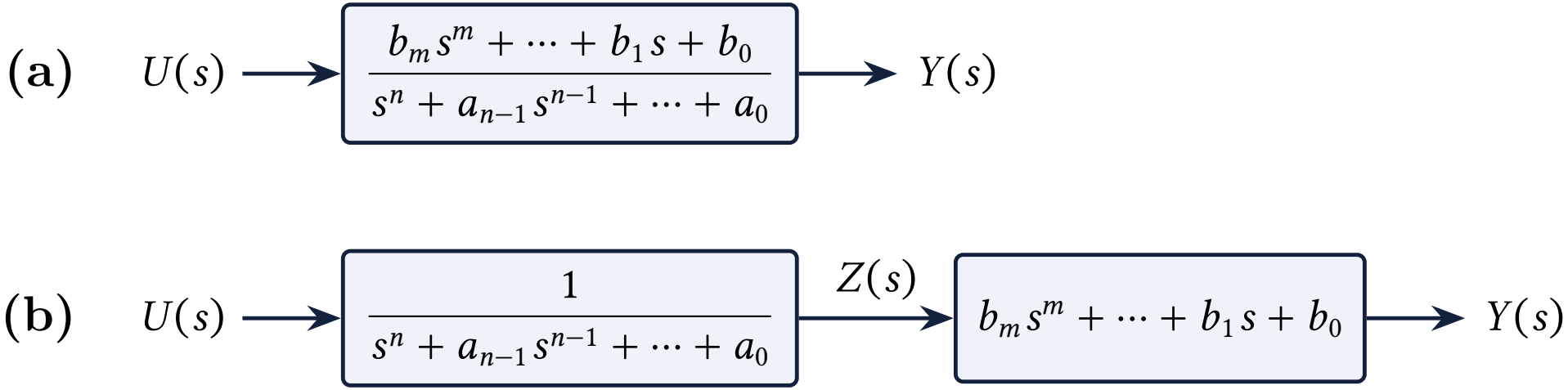

(a)
$U(s)$
$\frac{b_m s^m + \cdots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \cdots + a_0}$
$Y(s)$
(b)
$U(s)$
$\frac{1}{s^n + a_{n-1} s^{n-1} + \cdots + a_0}$
$Z(s)$
$b_m s^m + \cdots + b_1 s + b_0$
$Y(s)$